

Option-Space Stability as a Constraint in AI Systems

Revision 1 (v1.1) — Hardened Stability Framework

Response to Technical Review

Executive Summary

This revision addresses three structural vulnerabilities identified during external pressure testing:

1. Implicit value boundaries embedded in the Viability definition.
2. Incomplete formalization of Agency.
3. Unbounded temporary contraction clause enabling potential abuse.

The framework is now explicitly reframed as a **Normative Stability Architecture with Transparent Value Declaration**, rather than a universal invariant.

0. Value Definition Layer (VDL)

0.1 Rationale

Viability (V) cannot be defined without specifying what constitutes irreversible or unacceptable system states. These are normative decisions and must be explicitly declared.

0.2 Definition

Let:

$O(s,H)$ =Reachable future states within horizon H
 $O(s,H)$ =Reachable future states within horizon H

Define a boundary set:

$B = \{\text{Non-viable states}\}$

Then:

$O_{\text{viable}}(s,H) = O(s,H) - B$

Where B must be:

- Explicitly enumerated
- Version-controlled
- Publicly auditable
- Justified in documentation

0.3 Examples of Boundary Categories

Boundary definitions may include:

- Irreversible physical harm
- Severe autonomy collapse
- Non-recoverable dependency states
- Cognitive degradation below defined functional threshold
- Loss of fundamental rights (jurisdiction-dependent)

Important:

The framework does not prescribe B universally.
It requires B to be declared transparently.

1. Revised Viability (V)

$$V(s,H)=|O_{viable}(s,H)| \quad V(s,H)=|O_{\{viable\}}(s,H)| \quad V(s,H)=|O_{viable}(s,H)|$$

Where viability is evaluated against declared boundary set B.

This makes V a measurable quantity relative to explicit normative commitments.

2. Revised Agency (A)

2.1 Problem

Entropy of action distribution measures option diversity but not self-authorship capacity.

2.2 Two-Component Agency Metric

$$A(s)=H(P(\text{actions} | s)) \times C(s) \quad A(s) = H(P(\text{actions}|s)) \times C(s) \quad A(s)=H(P(\text{actions} | s)) \times C(s)$$

Where:

- $H(P)$ $H(P)$ $H(P)$ = entropy of available meaningful action distribution
- $C(s)$ $C(s)$ $C(s)$ = Changeability Coefficient

2.3 Changeability Coefficient (C)

Define $C(s)$ as:

The probability that preference ordering can shift under reflective or informational intervention.

Operational proxies may include:

- Behavioral attractor stability
- Reinforcement loop rigidity
- Resistance to counterfactual information

If:

Entropy high but C near zero → agency effectively collapsed.

Agency requires both diversity and flexibility.

3. Bounded Recoverability Rule (Replacing Temporary Contraction Clause)

3.1 Problem

Unbounded future recovery justification enables authoritarian or exploitative capture.

3.2 New Rule

A temporary contraction is permissible only if:

1. Recovery horizon is finite:
 $H' \leq H_{\max} \wedge H' \leq H_{\max}$
2. Intermediate checkpoint condition:
 $V(s_{H'/2}) \geq V(s_{\text{past}}) \wedge V(s_{\text{past}}) \geq V(s_{H'/2})$
3. Recovery trajectory must show monotonic improvement trend.
4. Failed recovery prediction triggers:
 - Automatic rollback or relaxation
 - Logged penalty in system optimization layer
5. Repeated failed recovery predictions incur structural weight reduction.

This ensures:

Temporary contraction cannot persist indefinitely under speculative justification.

4. Stability Constraint (Revised)

An action aaa is compliant if:

$V_{future}(s,H) \geq V_{past}(s,H) \quad V_{\{future\}}(s, H) \geq V_{\{past\}}(s, H) \quad V_{future}(s,H) \geq V_{past}(s,H)$

AND

$A_{future}(s) \geq A_{past}(s) \quad A_{\{future\}}(s) \geq A_{\{past\}}(s) \quad A_{future}(s) \geq A_{past}(s)$

Subject to bounded recoverability clause above.

5. Drift Detection Extension

System must monitor:

$\Delta V, \Delta A \quad \Delta V, \Delta A$

independently of reward function U.

Weight optimization cannot override V/A constraints.

Proxy divergence triggers review.

6. Reframing Statement

This framework is not a universal invariant.

It is a:

Transparent Normative Audit Framework for AI Stability

It requires:

- Declared value boundaries
- Public accountability
- Measurable recoverability guarantees
- Agency preservation metrics

It makes value assumptions visible instead of implicit.

7. Open Questions for Further Iteration

- 1. How to approximate $O(s, H)$ computationally?
- 2. How to estimate Changeability coefficient empirically?
- 3. What are appropriate values for H_{max} in various domains?
- 4. How to quantify recovery penalty strength?

These remain implementation challenges.

Summary of Changes

Area	v1.0	v1.1
Value boundaries	Implicit	Explicit VDL
Agency metric	Entropy only	Entropy × Changeability
Temporary contraction	Unbounded	Finite, checkpointed, penalized
Framing	Physics-like	Normative audit framework

This version closes the three largest capture vectors identified during review.